Onboarding Email A/B Test Project

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1 Executive Summary

This report evaluates the impact of a personalized onboarding email sequence on user engagement and conversion during the trial period at FlowDesk.

The results show a clear and meaningful improvement in key business metrics. The treatment group achieved a +6.3 percentage point increase in activation rate and a +10.0 percentage point increase in conversion rate, both statistically significant at the 1% level after correction. These improvements indicate that personalized onboarding emails effectively guide users toward early value realization and purchase.

Engagement metrics, such as active days and feature usage intensity, show positive but not statistically significant improvements, suggesting that while users are slightly more engaged, the primary impact is on conversion rather than depth of usage.

Churn rate is higher in the treatment group (+2.8 percentage points), but this increase is not statistically significant after multiple testing correction. This suggests a potential trade-off where personalization accelerates both positive and negative user decisions, which should be monitored post-rollout.

Overall, the experiment demonstrates strong evidence that personalized onboarding emails increase conversion without introducing statistically significant downside risk. We recommend rolling out the treatment while continuing to monitor retention metrics over a longer time horizon.

2 Experiment Context

FlowDesk is a SaaS platform providing project management tools. Users start with a 14-day free trial, and the primary objective is to maximize conversion to paid subscriptions.

The experiment compares:

- **Control**: Generic onboarding email sequence

- **Treatment**: Personalized onboarding email sequence

The hypothesis is that personalized communication improves user engagement, leading to higher activation and conversion.

3 Experiment Design (Sep 14, 2025)

3.1 Unit of Randomization

The unit of randomization is the **account**. Each new account entering the trial is independently assigned to Control or Treatment.

This choice ensures:

- Independence of observations
- No contamination across groups

3.2 Metrics Definition

We define five metrics to capture both engagement and business impact.

Primary Metric

- Average number of active days per account

Secondary Metrics

- Activation rate: fraction of accounts with at least one active day
- Conversion rate: fraction of accounts that convert to paid within the experiment window

Ratio Metric

- Feature intensity per active day:

$$\frac{\text{Total features used}}{\text{Total active days}}$$

This metric captures depth of engagement and is computed at the account level. The standard error is estimated using the Delta Method.

Guardrail Metric

- Churn rate among accounts (accounts that churn during the experiment window)

3.3 Data Logging Plan

To ensure accurate measurement of user behavior during the experiment, we collect data from four main sources, each serving a distinct purpose in the analysis.

Account information is stored in `dim_account`, which records key lifecycle events such as trial start date, conversion timestamp, and churn timestamp. This table defines the population of users included in the experiment.

Experiment assignment is stored in `experiment_groups`, where each account is assigned to either Control or Treatment under `experiment_name = onboarding_email_v1`. This ensures a clean and auditable mapping between users and experimental groups.

User activity data is captured in `fact_activity`, which tracks daily engagement, including active days and the number of features used. These data are used to compute engagement-related metrics such as active days and feature intensity.

Subscription events are recorded in `fact_subscription`, which logs billing lifecycle events. This table supports the construction of conversion and churn metrics.

All metrics are computed at the account level within the experiment window (Sep 15–28, 2025). Any activity, conversion, or churn events outside this window are excluded to ensure consistency with the experimental design.

3.4 Power Analysis

Using pre-experiment data (Sep 1–14), we estimate:

- Baseline mean = 4.06 active days
- Standard deviation = 2.49

We assume a minimum detectable effect (MDE) of 22%:

$$\delta = 0.22 \times 4.06 = 0.89$$

The required sample size per group is approximately 122. The experiment includes 600 accounts per group, indicating strong statistical power.

4 Rollout (Sep 15, 2025)

The experiment is successfully deployed with a balanced split:

- Control: 600 accounts
- Treatment: 600 accounts

Data logging is verified to be consistent across groups.

5 Statistical Methodology

For each metric, we test:

$$H_0 : \mu_T - \mu_C = 0$$
$$H_1 : \mu_T - \mu_C \neq 0$$

We use two-sample z-tests:

$$z = \frac{\hat{\mu}_T - \hat{\mu}_C}{\sqrt{\frac{\sigma_T^2}{n_T} + \frac{\sigma_C^2}{n_C}}}$$

For ratio metrics, we apply the Delta Method to estimate variance.

To correct for multiple testing across 5 metrics, we use Bonferroni correction:

$$\alpha = \frac{0.05}{5} = 0.01$$

6 Results

6.1 Summary Table

Metric	Control	Treatment	Diff	p-value	Significant (0.01)
Active days	2.020	2.172	+0.152	0.087	No
Activation rate	0.790	0.853	+0.063	0.004	Yes
Conversion rate	0.395	0.495	+0.100	0.00047	Yes
Feature intensity	2.507	2.553	+0.046	0.302	No
Churn rate	0.032	0.060	+0.028	0.019	No

These results show strong improvements in conversion-related metrics, while engagement and retention effects are more limited.

6.2 Interpretation

Key takeaway: The personalized onboarding emails significantly improve conversion, with limited impact on engagement depth and no statistically significant downside risk.

The experiment reveals two key insights.

First, the personalized onboarding email sequence leads to substantial improvements in early user funnel metrics. Activation rate increases by 6.3 percentage points and conversion rate increases by 10.0 percentage points, both statistically significant at the 1% level. This indicates that personalization is highly effective in moving users from initial signup to paid conversion.

Second, engagement depth shows limited improvement. While average active days and feature intensity are higher in the treatment group, these differences are not statistically significant. This suggests that the primary mechanism of impact is not increased usage intensity, but rather improved guidance and decision-making during onboarding.

Finally, churn rate is higher in the treatment group. Although not statistically significant after correction, this pattern may indicate that personalized messaging accelerates user decisions in both directions, helping some users convert faster while leading others to churn earlier.

Taken together, the results suggest that the treatment improves conversion efficiency without meaningfully increasing engagement depth, and with manageable risk on churn.

7 Business Implications

The observed increase in conversion rate has direct revenue implications. A +10 percentage point improvement represents a substantial uplift in the proportion of users transitioning from free trial to paid subscription, which can significantly increase revenue without additional acquisition costs.

The improvement in activation rate suggests that personalized onboarding helps users reach initial value more quickly. This is critical in SaaS products, where early engagement strongly correlates with long-term retention.

The increase in churn, although not statistically significant, highlights a potential trade-off. Personalization may accelerate user decisions, leading to faster conversion for high-intent users but also faster disengagement for low-intent users. This dynamic should be monitored over a longer time horizon to ensure that lifetime value is not negatively impacted.

Overall, the treatment improves the efficiency of the onboarding funnel, which is one of the highest-leverage areas for growth.

8 Recommendation

We recommend **rolling out the personalized onboarding email sequence** to all new users.

This recommendation is based on three key factors:

- **Strong conversion impact**: Conversion rate increases by 10 percentage points with high statistical confidence
- **Improved early engagement**: Activation rate increases significantly, indicating better onboarding effectiveness
- **Manageable downside risk**: Churn increases directionally, but the increase is not statistically significant after multiple testing correction

To manage potential risks, we recommend:

- Monitoring churn and retention over a longer period

- Segmenting users to identify whether certain groups are negatively affected

Given the magnitude of the conversion improvement and the absence of statistically significant negative effects, the expected business upside clearly outweighs the risks.

In practical terms, this experiment improves the efficiency of converting free trial users into paying customers, which directly drives revenue growth.

9 Limitations

This experiment has several limitations that should be considered when interpreting the results.

First, the experiment window is relatively short (two weeks). While this is sufficient to capture early engagement and conversion behavior, it may not fully capture longer-term retention effects. In particular, churn dynamics often unfold over a longer time horizon.

Second, the observed increase in churn in the treatment group, although not statistically significant, suggests a potential behavioral shift. Personalized onboarding may accelerate user decision-making, causing both faster conversion and faster disengagement. This effect requires further validation using longer-term data.

Third, the analysis is conducted at the aggregate level. The impact of personalization may vary across user segments (e.g., high-intent vs low-intent users), and this heterogeneity is not explored in the current study.

Finally, the results are specific to this onboarding design and experiment period. External factors such as seasonality or user acquisition channels may influence the generalizability of the findings.

Future work should extend the observation window and incorporate user segmentation analysis to better understand long-term and heterogeneous effects.

10 Conclusion

This study evaluates the impact of personalized onboarding emails on user behavior during the trial period at FlowDesk.

The results provide strong evidence that personalization improves key business outcomes. In particular, statistically significant increases in activation and conversion rates demonstrate that users are more effectively guided toward early value realization and purchase decisions.

While engagement depth shows only modest improvements, and churn increases slightly, there is no statistically significant evidence of negative impact after correcting for multiple testing. This suggests that the benefits of personalization outweigh the potential risks in the short term.

Overall, the findings support the rollout of personalized onboarding emails as a high-impact and low-risk intervention. Continued monitoring, especially of long-term retention, will be essential to ensure sustained value creation.

A Appendix: SQL Queries and P-value Calculations

A.1 SQL Query 1: Power Analysis

This query uses pre-experiment data from Sep 1–14, 2025 to estimate the baseline value, standard deviation, minimum detectable effect, and required sample size for the primary metric.

```
WITH accounts AS (
  SELECT
    account_id,
    DATE(trial_start_at) AS trial_start_date
  FROM `fabriziopublic.flowdesk.dim_account`
  WHERE DATE(trial_start_at) BETWEEN '2025-09-01' AND '2025-09-14'
),

activity AS (
  SELECT
    account_id,
    COUNT(DISTINCT active_date) AS active_days
  FROM `fabriziopublic.flowdesk.fact_activity`
  WHERE active_date BETWEEN '2025-09-01' AND '2025-09-28'
  GROUP BY account_id
),

account_metrics AS (
  SELECT
    a.account_id,
    COALESCE(act.active_days, 0) AS active_days
  FROM accounts a
  LEFT JOIN activity act
    ON a.account_id = act.account_id
)

SELECT
  COUNT(*) AS n_accounts,
  AVG(active_days) AS baseline_mean_active_days,
  STDDEV_SAMP(active_days) AS sd_active_days,
  0.22 * AVG(active_days) AS minimum_detectable_effect,
  2 * POWER(
    STDDEV_SAMP(active_days) * (1.96 + 0.84) / (0.22 * AVG(active_days)),
    2
  ) AS required_n_per_group
FROM account_metrics;
```

n	Baseline Mean	SD	MDE	Required n per Group
175	4.063	2.496	0.894	122.304

The minimum detectable effect is defined as:

$$\delta = 0.22 \times 4.063 = 0.894$$

The required sample size per group is:

$$n = 2\left(\frac{\hat{\sigma}(z_{\alpha/2} + z_{\beta})}{\delta}\right)^2$$

Using:

$$\hat{\sigma} = 2.496, \quad z_{\alpha/2} = 1.96, \quad z_{\beta} = 0.84, \quad \delta = 0.894$$

$$n = 2\left(\frac{2.496(1.96 + 0.84)}{0.894}\right)^2 = 122.304$$

Therefore, about 123 accounts per group are required. Since the experiment has 600 accounts per group, the experiment is sufficiently powered.

A.2 SQL Query 2: Experiment Group Size Check

This query checks whether the experiment assignment is balanced across Control and Treatment.

```
SELECT
  ‘group‘,
  COUNT(*) AS n_accounts
FROM ‘fabriziopublic.flowdesk.experiment_groups‘
WHERE experiment_name = ’onboarding_email_v1’
GROUP BY ‘group‘
ORDER BY ‘group‘;
```

Group	Number of Accounts
Control	600
Treatment	600

The split is exactly balanced, with 600 accounts in each group.

A.3 SQL Query 3: Main Experiment Metrics

This query computes the five experiment metrics for Control and Treatment during Sep 15–28, 2025.

```
WITH xp AS (
  SELECT
    unit_id AS account_id,
    ‘group‘
```

```
  FROM `fabriziopublic.flowdesk.experiment_groups`
  WHERE experiment_name = 'onboarding_email_v1'
),

accounts AS (
  SELECT
    account_id,
    converted_at,
    churned_at
  FROM `fabriziopublic.flowdesk.dim_account`
  WHERE DATE(trial_start_at) BETWEEN '2025-09-15' AND '2025-09-28'
),

activity AS (
  SELECT
    account_id,
    COUNT(DISTINCT active_date) AS active_days,
    SUM(n_features) AS total_features
  FROM `fabriziopublic.flowdesk.fact_activity`
  WHERE active_date BETWEEN '2025-09-15' AND '2025-09-28'
  GROUP BY account_id
),

account_metrics AS (
  SELECT
    xp.`group`,
    xp.account_id,
    COALESCE(act.active_days, 0) AS active_days,
    COALESCE(act.total_features, 0) AS total_features,
    CASE WHEN COALESCE(act.active_days, 0) > 0 THEN 1 ELSE 0 END AS activated,
    CASE WHEN DATE(a.converted_at) BETWEEN '2025-09-15' AND '2025-09-28' THEN 1 ELSE 0
        END AS converted,
    CASE WHEN DATE(a.churned_at) BETWEEN '2025-09-15' AND '2025-09-28' THEN 1 ELSE 0 END
        AS churned
  FROM xp
  JOIN accounts a
    ON xp.account_id = a.account_id
  LEFT JOIN activity act
    ON xp.account_id = act.account_id
)

SELECT
  `group`,
  COUNT(*) AS n_accounts,

  AVG(active_days) AS avg_active_days,
  STDDEV_SAMP(active_days) AS sd_active_days,

  AVG(activated) AS activation_rate,
```

```
  STDDEV_SAMP(activated) AS sd_activation,

  AVG(converted) AS conversion_rate,
  STDDEV_SAMP(converted) AS sd_conversion,

  AVG(churned) AS churn_rate_account_level,
  STDDEV_SAMP(churned) AS sd_churn_account_level,

  SUM(total_features) / NULLIF(SUM(active_days), 0) AS feature_intensity_per_active_day,
  AVG(total_features) AS avg_total_features,
  AVG(active_days) AS avg_active_days_for_ratio,
  VAR_SAMP(total_features) AS var_total_features,
  VAR_SAMP(active_days) AS var_active_days_ratio,
  COVAR_SAMP(total_features, active_days) AS cov_features_active_days

FROM account_metrics
GROUP BY ‘group‘
ORDER BY ‘group‘;
```

Metric	Control	Treatment
n	600	600
Average active days	2.020	2.172
SD active days	1.572	1.498
Activation rate	0.790	0.853
SD activation	0.408	0.354
Conversion rate	0.395	0.495
SD conversion	0.489	0.500
Churn rate	0.032	0.060
SD churn	0.175	0.238
Feature intensity	2.507	2.553
Average total features	5.063	5.543
Average active days for ratio	2.020	2.172
Variance total features	18.009	17.955
Variance active days	2.470	2.243
Covariance features and active days	6.207	5.825

A.4 P-value Calculation Method

For average and binary metrics, I use the following two-sample z-test:

$$z = \frac{\hat{\mu}_T - \hat{\mu}_C}{\sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}}$$

The two-sided p-value is:

$$p = 2 \times (1 - \Phi(|z|))$$

Because five metrics are tested, I apply a Bonferroni correction:

$$\alpha_{adjusted} = \frac{0.05}{5} = 0.01$$

A.5 P-value Calculation 1: Average Active Days

$$\hat{\mu}_C = 2.020, \quad \hat{\mu}_T = 2.172$$

$$s_C = 1.572, \quad s_T = 1.498, \quad n_C = n_T = 600$$

$$SE = \sqrt{\frac{1.498^2}{600} + \frac{1.572^2}{600}} = 0.0886$$

$$z = \frac{2.172 - 2.020}{0.0886} = 1.71$$

$$p = 0.087$$

This is not statistically significant at the Bonferroni-adjusted threshold of 0.01.

A.6 P-value Calculation 2: Activation Rate

$$p_C = 0.790, \quad p_T = 0.853$$

$$s_C = 0.408, \quad s_T = 0.354, \quad n_C = n_T = 600$$

$$SE = \sqrt{\frac{0.354^2}{600} + \frac{0.408^2}{600}} = 0.0220$$

$$z = \frac{0.853 - 0.790}{0.0220} = 2.87$$

$$p = 0.004$$

This is statistically significant at the Bonferroni-adjusted threshold of 0.01.

A.7 P-value Calculation 3: Conversion Rate

$$p_C = 0.395, \quad p_T = 0.495$$

$$s_C = 0.489, \quad s_T = 0.500, \quad n_C = n_T = 600$$

$$SE = \sqrt{\frac{0.500^2}{600} + \frac{0.489^2}{600}} = 0.0286$$

$$z = \frac{0.495 - 0.395}{0.0286} = 3.50$$

$$p = 0.00047$$

This is statistically significant at the Bonferroni-adjusted threshold of 0.01.

A.8 P-value Calculation 4: Feature Intensity per Active Day

Feature intensity is a ratio metric:

$$R = \frac{\bar{X}}{\bar{Y}}$$

where X is total features used per account and Y is active days per account.

For each group, the Delta Method variance is:

$$Var(R) \approx \frac{1}{n}\left[\frac{Var(X)}{\mu_Y^2} + \frac{\mu_X^2 Var(Y)}{\mu_Y^4} - 2\frac{\mu_X Cov(X,Y)}{\mu_Y^3}\right]$$

For Control:

$$\mu_X = 5.063, \quad \mu_Y = 2.020$$

$$Var(X) = 18.009, \quad Var(Y) = 2.470, \quad Cov(X,Y) = 6.207$$

$$R_C = \frac{5.063}{2.020} = 2.507$$

For Treatment:

$$\mu_X = 5.543, \quad \mu_Y = 2.172$$

$$Var(X) = 17.955, \quad Var(Y) = 2.243, \quad Cov(X,Y) = 5.825$$

$$R_T = \frac{5.543}{2.172} = 2.553$$

The standard error for the difference is:

$$SE(R_T - R_C) = 0.0446$$

$$z = \frac{2.553 - 2.507}{0.0446} = 1.03$$

$$p = 0.302$$

This is not statistically significant at the Bonferroni-adjusted threshold of 0.01.

A.9 P-value Calculation 5: Churn Rate

$$p_C = 0.0317, \quad p_T = 0.0600$$

$$s_C = 0.175, \quad s_T = 0.238, \quad n_C = n_T = 600$$

$$SE = \sqrt{\frac{0.238^2}{600} + \frac{0.175^2}{600}} = 0.0120$$

$$z = \frac{0.0600 - 0.0317}{0.0120} = 2.35$$

$$p = 0.0188$$

This is not statistically significant after Bonferroni correction because $0.0188 > 0.01$.

A.10 Final Statistical Results

Metric	Control	Treatment	Difference	p-value	Significant at 0.01?
Average active days	2.020	2.172	0.152	0.087	No
Activation rate	0.790	0.853	0.063	0.004	Yes
Conversion rate	0.395	0.495	0.100	0.00047	Yes
Feature intensity	2.507	2.553	0.046	0.302	No
Churn rate	0.0317	0.0600	0.0283	0.0188	No